

## SAFETY DATA SHEET

Creation Date 05-May-2009

Revision Date 27-March-2024

Revision Number 5

### 1. Identification

**Product Name** 1,4-Dioxane

**Cat No. :** 22913

**CAS-No** 123-91-1  
**Synonyms** Diox

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                   |             |
|-------------------------------------------------------------------|-------------|
| <b>Flammable liquids</b>                                          | Category 2  |
| <b>Serious Eye Damage/Eye Irritation</b>                          | Category 2  |
| <b>Carcinogenicity</b>                                            | Category 1B |
| <b>Specific target organ toxicity (single exposure)</b>           | Category 3  |
| Target Organs - Respiratory system, Central nervous system (CNS). |             |
| <b>Physical Hazards Not Otherwise Classified</b>                  | Category 1  |
| May form explosive peroxides                                      |             |
| <b>Health Hazards Not Otherwise Classified</b>                    | Category 1  |
| Repeated exposure may cause skin dryness or cracking              |             |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Highly flammable liquid and vapor  
Causes serious eye irritation  
May cause respiratory irritation  
May cause cancer  
May form explosive peroxides  
Repeated exposure may cause skin dryness or cracking

**Precautionary Statements****Prevention**

Keep container tightly closed  
Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Use non-sparking tools  
Take action to prevent static discharges

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF exposed or concerned: Get medical advice/attention  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Contains a known or suspected endocrine disruptor

### 3. Composition/Information on Ingredients

| Component   | CAS-No   | Weight % |
|-------------|----------|----------|
| 1,4-Dioxane | 123-91-1 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get

|                                        |                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                        | medical attention.                                                                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.     |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.          |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water.                                                          |
| <b>Most important symptoms/effects</b> | . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                 |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | Water may be ineffective                                                                                                                |
| <b>Flash Point</b>                      | 12 °C / 53.6 °F                                                                                                                         |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 355 °C / 671 °F                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 22%                                                                                                                                     |
| <b>Lower</b>                            | 2%                                                                                                                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. May form explosive peroxides. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). peroxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                  |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. To avoid ignition of |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Flammables area. May form explosive peroxides. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Protect from moisture. Incompatible Materials. Strong oxidizing agents. Reducing Agent. Halogens.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component   | Alberta                                          | British Columbia    | Ontario TWAEV       | Quebec                                           | ACGIH TLV           | OSHA PEL                                                                                                           | NIOSH                                                             |
|-------------|--------------------------------------------------|---------------------|---------------------|--------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 1,4-Dioxane | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm<br>Skin | TWA: 20 ppm<br>Skin | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm<br>Skin | (Vacated) TWA: 25 ppm<br>(Vacated) TWA: 90 mg/m <sup>3</sup><br>Skin<br>TWA: 100 ppm<br>TWA: 360 mg/m <sup>3</sup> | IDLH: 500 ppm<br>Ceiling: 1 ppm<br>Ceiling: 3.6 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Tight sealing safety goggles Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.7 mm          | As tested under EN374-3                                |
| Viton (R)      | > 480 minutes     | 0.7 mm          | Determination of Resistance to Permeation by Chemicals |
|                |                   |                 | Permeation rate 38 µg/cm <sup>2</sup> /min             |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

#### **Environmental exposure controls**

No information available.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Liquid                       |
| Appearance                             | Colorless                    |
| Odor                                   | Petroleum distillates        |
| Odor Threshold                         | No information available     |
| pH                                     | 6-8 500 g/l aq.sol           |
| Melting Point/Range                    | 12 °C / 53.6 °F              |
| Boiling Point/Range                    | 101 °C / 213.8 °F @ 760 mmHg |
| Flash Point                            | 12 °C / 53.6 °F              |
| Evaporation Rate                       | No information available     |
| Flammability (solid,gas)               | Not applicable               |
| Flammability or explosive limits       |                              |
| Upper                                  | 22%                          |
| Lower                                  | 2%                           |
| Vapor Pressure                         | 41 mbar @ 20 °C              |
| Vapor Density                          | 3                            |
| Specific Gravity                       | 1.034                        |
| Solubility                             | Soluble in water             |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | 355 °C / 671 °F              |
| Decomposition Temperature              | No information available     |
| Viscosity                              | 1.32 mPa.s @ 20 °C           |
| Molecular Formula                      | C4 H8 O2                     |
| Molecular Weight                       | 88.11                        |

### 10. Stability and reactivity

|                                  |                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                                                                                            |
| Stability                        | May form explosive peroxides. Hygroscopic.                                                                                                                                                            |
| Conditions to Avoid              | Incompatible products. Heat, flames and sparks. Exposure to air or moisture over prolonged periods. Keep away from open flames, hot surfaces and sources of ignition. Exposure to moist air or water. |
| Incompatible Materials           | Strong oxidizing agents, Reducing Agent, Halogens                                                                                                                                                     |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), peroxides                                                                                                                                    |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                                                              |
| Hazardous Reactions              | None under normal processing.                                                                                                                                                                         |

### 11. Toxicological information

#### **Acute Toxicity**

**Product Information**  
**Component Information**

| Component   | LD50 Oral                                | LD50 Dermal                  | LC50 Inhalation       |
|-------------|------------------------------------------|------------------------------|-----------------------|
| 1,4-Dioxane | 5170 mg/kg ( Rat )<br>4200 mg/kg ( Rat ) | LD50 = 7600 mg/kg ( Rabbit ) | 48.5 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** Acetonitrile; Tetrachloroethylene

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**
**Irritation** Irritating to eyes, respiratory system and skin

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component   | CAS-No   | IARC     | NTP                    | ACGIH | OSHA | Mexico |
|-------------|----------|----------|------------------------|-------|------|--------|
| 1,4-Dioxane | 123-91-1 | Group 2B | Reasonably Anticipated | A3    | X    | A3     |

*IARC (International Agency for Research on Cancer)*
*IARC (International Agency for Research on Cancer)*
*Group 1 - Carcinogenic to Humans*
*Group 2A - Probably Carcinogenic to Humans*
*Group 2B - Possibly Carcinogenic to Humans*
*NTP: (National Toxicity Program)*
*NTP: (National Toxicity Program)*
*Known - Known Carcinogen*
*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*
*ACGIH: (American Conference of Governmental Industrial Hygienists)*
*A1 - Known Human Carcinogen*
*A2 - Suspected Human Carcinogen*
*A3 - Animal Carcinogen*
*ACGIH: (American Conference of Governmental Industrial Hygienists)*
*Mexico - Occupational Exposure Limits - Carcinogens*
*Mexico - Occupational Exposure Limits - Carcinogens*
*A1 - Confirmed Human Carcinogen*
*A2 - Suspected Human Carcinogen*
*A3 - Confirmed Animal Carcinogen*
*A4 - Not Classifiable as a Human Carcinogen*
*A5 - Not Suspected as a Human Carcinogen*
**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information**
**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

| Component   | Freshwater Algae | Freshwater Fish                                 | Microtox                                        | Water Flea          |
|-------------|------------------|-------------------------------------------------|-------------------------------------------------|---------------------|
| 1,4-Dioxane | Not listed       | LC50: = 9850 mg/L, 96h<br>(Pimephales promelas) | EC50 = 610 mg/L 5 min<br>EC50 = 668 mg/L 15 min | EC50 = 163 mg/L 48h |

|  |  |                                                                                                                                                                                                                                              |                        |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--|
|  |  | LC50: 10306 - 14742 mg/L, 96h static (Pimephales promelas)<br>LC50: = 9850 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 10000 mg/L, 96h semi-static (Lepomis macrochirus)<br>LC50: > 10000 mg/L, 96h static (Lepomis macrochirus) | EC50 = 733 mg/L 30 min |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--|

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component   | log Pow |
|-------------|---------|
| 1,4-Dioxane | -0.42   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component              | RCRA - U Series Wastes | RCRA - P Series Wastes |
|------------------------|------------------------|------------------------|
| 1,4-Dioxane - 123-91-1 | U108                   | -                      |

### 14. Transport information

#### DOT

UN-No UN1165  
Proper Shipping Name DIOXANE  
Hazard Class 3  
Packing Group II

#### TDG

UN-No UN1165  
Proper Shipping Name DIOXANE  
Hazard Class 3  
Packing Group II

#### IATA

UN-No UN1165  
Proper Shipping Name DIOXANE  
Hazard Class 3  
Packing Group II

#### IMDG/IMO

UN-No UN1165  
Proper Shipping Name DIOXANE  
Hazard Class 3  
Packing Group II

### 15. Regulatory information

#### International Inventories

| Component   | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1,4-Dioxane | 123-91-1 | X   | -    | X    | ACTIVE                                        | 204-661-8 | -      | -   |

| Component | CAS-No | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|--------|-------|------|------|------|------|------|-------|-------|
|-----------|--------|-------|------|------|------|------|------|-------|-------|

|             |          |   |          |   |   |   |   |   |   |
|-------------|----------|---|----------|---|---|---|---|---|---|
| 1,4-Dioxane | 123-91-1 | X | KE-10463 | X | X | X | X | X | X |
|-------------|----------|---|----------|---|---|---|---|---|---|

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component   | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA)         |
|-------------|------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|
| 1,4-Dioxane | Part 1, Group A Substance Part 4 Substance           |                                                                            | Subject to Monitoring and Surveillance Activities |

**Legend**

NPRI - National Pollutant Release Inventory

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)                                                                                                                                                                                          |
|-------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,4-Dioxane | -                                                                   | Use restricted. See item 75. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details) | SVHC Candidate list - 204-661-8 - Carcinogenic (Article 57a)<br><br>Equivalent level of concern having probable serious effects to the environment (Article 57f - environment)<br><br>Equivalent level of concern having probable serious effects to human health (Article 57f - human health) |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

**REACH links**<https://echa.europa.eu/substances-restricted-under-reach><https://echa.europa.eu/candidate-list-table><https://echa.europa.eu/authorisation-list>**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component   | CAS-No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| 1,4-Dioxane | 123-91-1 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

  

| Component | CAS-No | Seveso III Directive (2012/18/EC) - Qualifying Quantities | Seveso III Directive (2012/18/EC) - Qualifying Quantities | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|--------|-----------------------------------------------------------|-----------------------------------------------------------|----------------------------|------------------------------------|
|           |        |                                                           |                                                           |                            |                                    |

|             |          | for Major Accident Notification | for Safety Report Requirements |                |                |
|-------------|----------|---------------------------------|--------------------------------|----------------|----------------|
| 1,4-Dioxane | 123-91-1 | Not applicable                  | Not applicable                 | Not applicable | Not applicable |

## 16. Other information

**Prepared By**

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date**

05-May-2009

**Revision Date**

27-March-2024

**Print Date**

27-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**